Cassandra Tong Ye

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Research Statement

I am a 3rd year PhD student advised by Kristina Monakhova in the Cornell Computational Imaging Lab. My research focuses on uncertainty quantification and trustworthy machine learning for imaging inverse problems. I develop statistically grounded, data-driven methods for assessing and calibrating uncertainty in learned models, with the broader goal of enabling robust and reliable inference in medical imaging and scientific discovery.

Education

Cornell University 2024 – Present
Ph.D. in Computer Science, Advisor: Kristina Monakhova

Massachusetts Institute of Technology 2020 – 2024
B.S. in Computer Science and Engineering, Advisor: Sixian You

Industry Experience

Adobe Summer 2026
Machine Learning Intern on the Nextcam Team, led by Marc Levoy

- Researched efficient ways to estimate epistemic uncertainty in high-dimensional image restoration problems.

Microsoft Spring 2022
Program Manager Intern on the iOS Devices Team

Publications

- *QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems*
  C.T. Ye, S. Li, T. King, K. Monakhova. ***NeurIPS, 2026.***
- *Learned, uncertainty-driven adaptive acquisition for photon-efficient scanning microscopy*
  C.T. Ye, J. Han, K. Liu, A. Angelopoulos, L. Griffith, K. Monakhova, S. You. ***Optics Express, 33(6):12269, 2025.***
- *Exact-match search with functional variant prediction enables automated DNA screening*
  D. Gretton, B. Wang, R. Edison, L. Foner, J. Berlips, T. Vogel, M. Kysel, W. Chen, F. Sage-Ling, L. Van Hauwe, ...**C.T. Ye**, K. Uberoy, R. Rivest, M. Gao, Y. Yu, C. Baum, I. Damgard, A.C. Yao, K.M. Esvelt. ***bioRxiv, 2024.03.20.585782, 2024.***

Workshops and Presentations

- *Quantifying Uncertainty with MR-PhICC: MRI Reconstruction with Physics-Informed Conformal Calibration for Imaging Inverse Problems*
  Nora Gera, **C.T. Ye**, Kristina Monakhova.
  Innovation to Implementation in Imaging (i2i) Workshop, 2026.
- *QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems*
  C.T. Ye, S. Li, T. King, K. Monakhova.
  International Conference on Computational Photography (ICCP), 2025, 2026.
- *Quantile UQNet: A Conformal, Nonlinear Scaling Model for Uncertainty Prediction*
  C.T. Ye, Tyler King, Kristina Monakhova.
  Spotlight Presentation, **Uncertainty Quantification for Computer Vision (UnCV) Workshop at CVPR**, 2025.
- *Learned, Uncertainty-driven adaptive acquisition for photon-efficient scanning microscopy*
  C.T. Ye, J. Han, K. Liu, A. Angelopoulos, L. Griffith, K. Monakhova, S. You.
  International Conference on Computational Photography (ICCP), 2024.
- *Leveraging uncertainty quantification in adaptive multiphoton microscopy acquisition*
  C.T. Ye, coauthors.
  Best Paper Award, **SPIE Photonics West**, 2024.

Invited Talks

- *Uncertainty Quantification and its Applications in Computational Imaging*
  Invited talks at Purdue University and Northwestern University, Nov. 2025. (Hosts: Prof. Sotiris Nousias, Prof. Emma Alexander)
- *QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems*
  Cornell Vision and Graphics Seminar, 2025.

Awards and Fellowships

- Best Paper Award, SPIE Photonics West, 2024.
- NSF GRFP Honorable Mention, 2024.
- MIT Sandbox Innovation Fund, 2023.
- MIT Artificial Intelligence Alignment Fellowship, 2023.
- MIT EECS Philips Undergraduate Research and Innovation Scholar, 2022–2023.

Teaching Experience and Service

- Mentor, Cornell New Vision Engineering (NVE), 2024–2025.
- Teaching Assistant, CS4670 Computer Vision, Cornell University, Spring 2025.
- Teaching Assistant, CS4620 Computer Graphics, Cornell University, Fall 2024.
- Laboratory Assistant, MIT Introduction to Technology, Engineering, and Science (MITES), Summer 2023.